

SSS 1-3 Controlled Experiment Garden • Practical Activity Cards

Progression: DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

PA-421 Frame a Causal Question	Complete the investigation using explicit assumptions, systematic measurements, and...
PA-422 Design a Factorial Experiment	Define the objective, boundary, variables, constraints, and performance criteria. D...
PA-423 Identify Main Effects	Complete the investigation using explicit assumptions, systematic measurements, and...
PA-424 Interpret an Interaction	Use quantitative evidence and stated criteria. Compare alternatives or conditions, ...
PA-425 Detect Confounding	Identify the abnormal signal or performance pattern, formulate plausible causes, te...
PA-426 Develop a Blinded Measurement Plan	Define the objective, boundary, variables, constraints, and performance criteria. D...
PA-427 Perform an Instrument Calibration	Use a documented protocol and appropriate instrument or reference. Repeat where nee...
PA-428 Quantify Repeatability	Use the specified measurements, equation, and assumptions. Show units throughout, c...
PA-429 Conduct a Sensitivity Analysis	Complete the investigation using explicit assumptions, systematic measurements, and...
PA-430 Evaluate Causal Evidence	Use quantitative evidence and stated criteria. Compare alternatives or conditions, ...